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Module 2.4 International Economics and Development

1. Exchange Rates

Exchange rate is the price of one currency expressed in terms of another currency. For example, £1 = $1.25 means one pound can be exchanged for 1.25 US dollars.

1.1 Exchange Rate Systems

Floating Exchange Rate

The exchange rate is determined by the forces of **supply and demand** in the foreign exchange market, with no government intervention. Most major currencies (pound, dollar, euro, yen) operate managed floats.

Fixed Exchange Rate

The government or central bank **pegs** the currency at a specific rate against another currency and intervenes (buying/selling currency, adjusting interest rates) to maintain the peg. Example: many Gulf states peg to the US dollar.

Managed Float

Primarily market-determined but the central bank intervenes occasionally to smooth excessive fluctuations or prevent misalignment. The UK operates a managed float.

1.2 Determination of a Floating Exchange Rate

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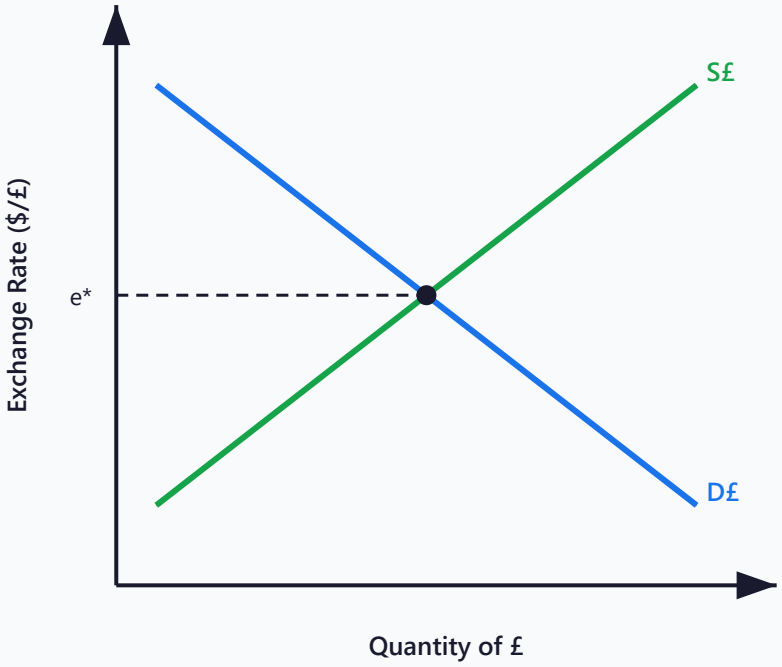

Figure 1: Exchange rate determination in a floating system — equilibrium where demand for and supply of pounds intersect

Demand for Pounds (D£) — comes from:

- Foreigners buying UK exports (they need pounds to pay)
- Foreign investors investing in the UK (buying UK assets, shares, bonds)
- Speculators expecting the pound to appreciate
- Tourists visiting the UK

Supply of Pounds (S£) — comes from:

- UK residents buying imports (selling pounds to buy foreign currency)
- UK investors investing abroad
- Speculators expecting the pound to depreciate
- UK tourists spending abroad

1.3 Appreciation and Depreciation

Appreciation: An increase in the value of a currency (e.g. £1 rises from $1.25 to $1.40). Each pound buys more foreign currency.
Depreciation: A decrease in the value of a currency (e.g. £1 falls from $1.25 to $1.10). Each pound buys less foreign currency.

Causes of Appreciation

- Higher UK interest rates (attract foreign capital inflows)
- Increased demand for UK exports
- Inward foreign direct investment
- Speculation that the pound will rise
- Higher relative economic growth attracting investment
- Lower UK inflation relative to trading partners

Causes of Depreciation

- Lower UK interest rates relative to other countries
- Current account deficit (more pounds sold to buy imports)
- Political uncertainty (e.g. Brexit referendum caused a sharp fall in sterling)
- Higher UK inflation making exports less competitive
- Speculation against the pound

1.4 Effects of Exchange Rate Changes

Effect	Appreciation (stronger £)	Depreciation (weaker £)
Exports	More expensive for foreigners — export volumes fall	Cheaper for foreigners — export volumes rise
Imports	Cheaper for UK consumers — import volumes rise	More expensive — import volumes fall
Current account	Tends to worsen (more imports, fewer exports)	Tends to improve (more exports, fewer imports)

Inflation	Reduces inflation (cheaper imports, lower costs)	Increases inflation (dearer imports, higher costs)
Growth	May slow growth (reduced export demand)	May boost growth (increased export demand)
Firms	Exporters suffer, importers benefit	Exporters benefit, importers suffer

1.5 The Marshall-Lerner Condition and J-Curve

Marshall-Lerner condition: A depreciation will only improve the current account if the sum of the price elasticities of demand for exports and imports is greater than 1 ($|PED_X| + |PED_M| > 1$).

The **J-Curve effect** explains why the current account may initially worsen following a depreciation before improving:

- **Short run:** Import and export contracts are fixed in advance. Import prices rise immediately (more expensive in pounds), but volumes do not change. Export volumes also do not change quickly. The trade balance worsens.
- **Long run:** As contracts are renegotiated, export volumes rise (now cheaper) and import volumes fall (now dearer). The current account improves — provided the Marshall-Lerner condition is met.

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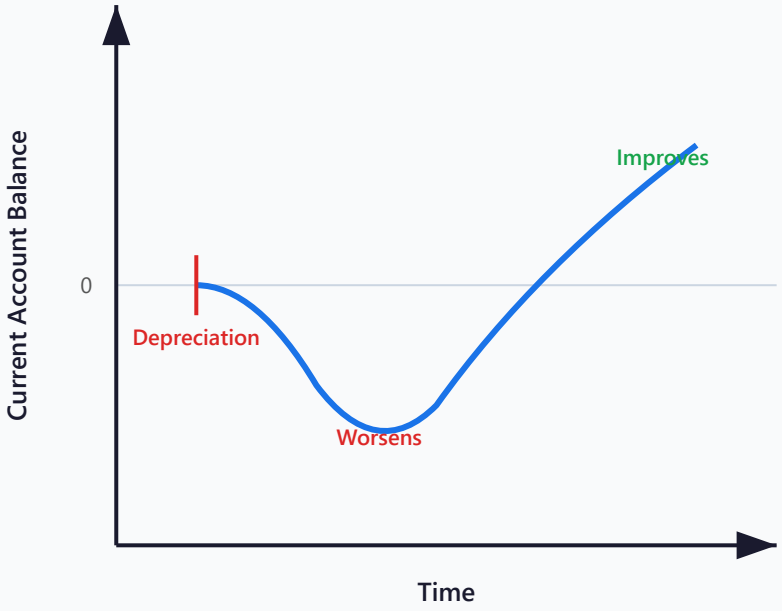

Figure 2: The J-Curve — current account initially worsens after depreciation before improving

Case Study: Sterling's Fall After the Brexit Referendum (2016)

Following the EU referendum result on 23 June 2016, the pound fell approximately 10% against the dollar (from $1.50 to $1.33) and 15% against the euro. This was the sharpest single-day fall in sterling's history, driven by speculation about economic uncertainty and reduced foreign investment. Effects: import prices rose sharply, contributing to CPI inflation rising from 0.5% to 3% by late 2017 (cost-push). UK exporters initially benefited from improved competitiveness, and the FTSE 100 rose (many listed companies earn revenue in dollars). However, the J-Curve effect was visible — the UK current account did not improve initially. Real wages fell as inflation outpaced wage growth, squeezing consumer living standards.

2. Trade Theory and Protectionism

2.1 The Case for Free Trade

Based on the principle of **comparative advantage** (see Module 2.1), free trade allows countries to specialise in goods where they have the lowest opportunity cost, increasing total global output and enabling mutual gains from trade.

- Lower prices for consumers (access to cheaper goods)
- Greater product variety
- Economies of scale from larger markets
- Increased competition drives efficiency and innovation
- Technology transfer through FDI and trade

2.2 Protectionism — Methods

Protectionism is the use of government policies to restrict free trade and protect domestic industries from foreign competition.

Method	How it Works	Effects
Tariff	A tax on imported goods, raising their price in the domestic market	Import price rises, domestic producers can compete, government earns revenue, but consumers pay higher prices
Quota	A physical limit on the quantity of imports allowed	Restricts supply, raises prices, no government revenue (unlike tariffs), may create shortages
Subsidy to domestic producers	Government payment to domestic firms, lowering their costs and making them more competitive against imports	Domestic output rises, imports fall, but opportunity cost of government spending
Non-tariff barriers	Regulations, standards, bureaucracy that make it harder to import (e.g. health and safety rules, labelling requirements, customs delays)	Harder to detect and challenge; can be disguised as legitimate regulation
Embargo	A complete ban on trade with a particular country or of a particular good	Used for political or security reasons (e.g. sanctions on Russia)

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2.3 Tariff Diagram

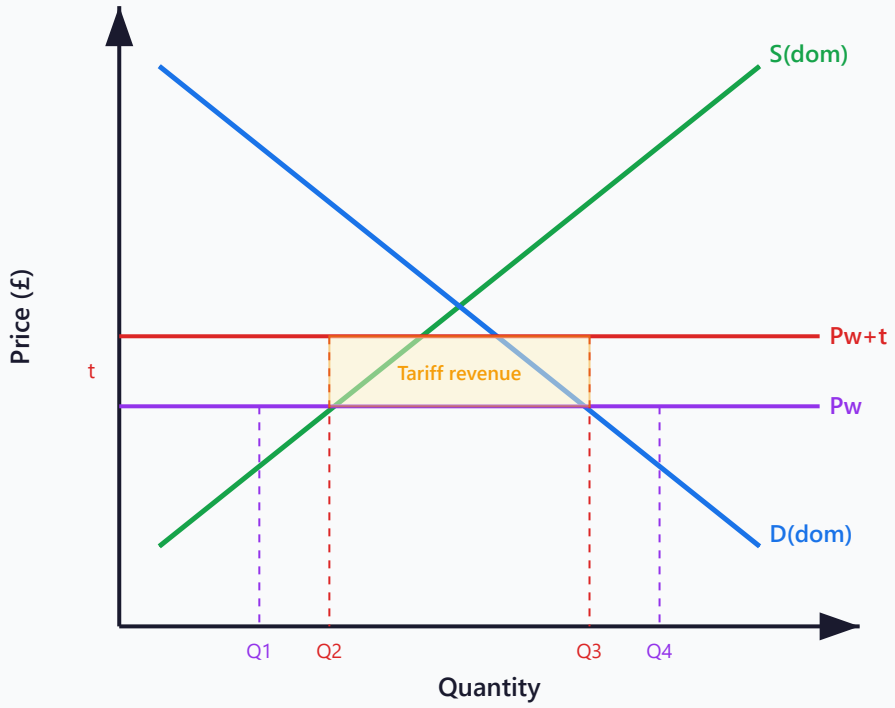

Figure 3: Effect of a tariff — raises domestic price from Pw to Pw+t, domestic supply rises (Q1 to Q2), domestic demand falls (Q4 to Q3), imports shrink

2.4 Arguments For Protectionism

- **Infant industry:** New industries may need temporary protection to develop economies of scale before competing internationally.
- **Protect employment:** Prevent job losses in industries facing cheap imports (e.g. steel, textiles).
- **Strategic industries:** Protect industries vital for national security (defence, food, energy).
- **Anti-dumping:** Prevent foreign firms selling below cost to destroy domestic competition.
- **Improve current account:** Reducing imports improves the trade balance.
- **Raise government revenue:** Tariffs generate tax income (important in developing countries with limited domestic tax bases).
- **Environmental/labour standards:** Prevent a "race to the bottom" where countries compete by lowering standards.

2.5 Arguments Against Protectionism

- **Higher prices for consumers** — tariffs and quotas raise prices and reduce choice.
- **Retaliation:** Trading partners may impose their own tariffs, reducing exports (trade wars).
- **Inefficiency:** Protected firms lack competitive pressure to become efficient (X-inefficiency).
- **Infant industries may never grow up** — protection can become permanent (lobbying to maintain it).
- **Misallocation of resources:** Resources stay in uncompetitive industries instead of moving to areas of comparative advantage.
- **Deadweight welfare loss:** Tariffs create welfare losses (reduced consumer surplus).

Case Study: US-China Trade War (2018-2020)

In 2018, the US imposed tariffs on approximately $360 billion of Chinese imports (up to 25% on many goods), citing unfair trade practices and intellectual property theft. China retaliated with tariffs on $110 billion of US goods (targeting agriculture and manufacturing). Effects: US consumer prices rose (estimated $831 per household per year in higher costs), US agricultural exports to China collapsed (soybean exports fell by 75%), global supply chains were disrupted, and business investment uncertainty increased. Studies estimated the tariffs reduced US GDP by 0.3-0.7%. The trade war demonstrated the costs of protectionism — both countries were worse off, though the tariffs did reduce the bilateral trade deficit somewhat.

3. Economic Development

3.1 Growth vs Development

Economic growth is an increase in real GDP — a quantitative measure of output.

Economic development is a broader concept encompassing improvements in living standards, health, education, freedom, equality, and well-being. A country can experience growth without development (e.g. if growth only benefits elites or comes at the expense of the environment).

3.2 Indicators of Development

Indicator	What it Measures	Strengths	Weaknesses
GDP per capita (PPP)	Average income adjusted for purchasing power	Widely available, allows comparison	Ignores distribution, informal economy, non-monetary welfare
HDI	Life expectancy + education + GNI per capita	Broader than GDP alone, captures health and education	Still an average (ignores inequality within countries), limited dimensions
Gini coefficient	Income inequality (0-1 scale)	Direct inequality measure	Does not capture wealth inequality or absolute poverty
Multidimensional Poverty Index (MPI)	Health, education, and living standards deprivations	Captures non-income dimensions of poverty	Data-intensive, not available for all countries
Literacy rate	% of adults who can read and write	Simple, important for development	Binary measure (literate/not) misses nuance
Life expectancy	Average years a newborn is expected to live	Good proxy for health and living conditions	National average hides regional/group variation

3.3 Barriers to Development

- **Primary product dependency:** Reliance on exporting raw materials (coffee, copper, oil) with volatile prices and declining terms of trade.
- **Savings gap:** Low incomes mean low savings, which means low investment and low growth — a vicious circle (Harrod-Domar model).

- **Corruption and weak institutions:** Poor governance, corruption, lack of rule of law deter investment and misallocate resources.
- **Debt burden:** Many developing countries carry large debts — interest payments divert resources from development spending.
- **Lack of infrastructure:** Poor transport, energy, water, and communications limit economic activity.
- **Population growth:** Rapid population growth can outstrip economic growth, reducing GDP per capita. It strains public services.
- **Human capital deficiency:** Poor education and healthcare reduce worker productivity and innovation.
- **Trade barriers:** Developed countries' agricultural subsidies and trade barriers restrict access for developing country exports.
- **Geography and climate:** Landlocked countries, disease-prone regions (malaria zones), climate vulnerability limit growth.
- **Capital flight:** Wealthy elites and MNCs may move profits out of the country, reducing domestic investment.

Case Study: The Resource Curse — Nigeria

Nigeria is Africa's largest oil producer and has earned trillions of dollars from petroleum exports since the 1970s. Yet approximately 40% of Nigerians live below the poverty line, infrastructure is poor, and the country ranks 163rd out of 191 on the HDI. This "resource curse" (or paradox of plenty) occurred because oil wealth fuelled corruption (estimated $400 billion in oil revenue stolen or wasted since independence), caused Dutch Disease (manufacturing and agriculture declined as the oil sector attracted resources and the currency appreciated), created dependency on a single volatile export, and concentrated wealth among a political elite. Nigeria exemplifies how economic growth (from oil revenue) can occur without broad-based economic development.

4. Development Strategies

4.1 Market-Oriented Strategies

Trade Liberalisation

Opening the economy to international trade based on comparative advantage. Joining the WTO, reducing tariffs, attracting FDI.

Advantages: Access to larger markets, technology transfer, competitive pressure, economies of scale.

Risks: Domestic industries destroyed by competition, dependency on developed-country markets, terms of trade may deteriorate.

Structural Adjustment Programmes (SAPs)

Conditions attached to IMF/World Bank loans requiring market-oriented reforms: privatisation, deregulation, trade liberalisation, fiscal austerity, currency devaluation.

Criticisms: SAPs often worsened poverty in the short term (spending cuts hit healthcare and education), undermined food sovereignty, and prioritised debt repayment over development.

Foreign Direct Investment (FDI)

Encouraging multinational corporations to invest — bringing capital, technology, management skills, and employment.

Advantages: Job creation, technology transfer, infrastructure improvement, export revenue.

Risks: Profit repatriation (money leaves the country), exploitation of cheap labour, environmental damage, MNCs may have more power than local government.

4.2 Interventionist Strategies

Import Substitution Industrialisation (ISI)

Protecting domestic industry from imports through tariffs and quotas, developing manufacturing capacity to replace imported goods.

Advantages: Develops domestic industry, reduces import dependence, creates employment.

Disadvantages: Protected firms may remain inefficient, limited domestic market restricts scale, consumers face higher prices, may provoke retaliation.

Export-Led Growth

Focusing on producing goods for export markets, using competitive wages and exchange rates to attract manufacturing.

Success stories: South Korea, Taiwan, Singapore, China — the "Asian Tigers" grew rapidly through export orientation.

Risks: Dependency on foreign demand, race to the bottom on wages, vulnerability to global recessions.

Investment in Human Capital

Government spending on education, healthcare, and skills training to improve labour productivity and long-run growth potential.

Industrial Policy

Government actively directing resources toward strategic industries through subsidies, state-owned enterprises, and planning.

Case Study: South Korea's Development Miracle

South Korea transformed from one of the poorest countries in the world in 1960 (GDP per capita similar to sub-Saharan Africa) to a high-income developed economy by the 2000s. Key strategies included: government-directed export-led industrialisation (supporting chaebol conglomerates like Samsung, Hyundai, LG), massive investment in education (South Korea went from 35% literacy in 1945 to near-universal literacy, and now has one of the world's highest rates of tertiary education), strategic use of tariff protection for infant industries followed by gradual liberalisation, high domestic savings rates channelled into investment, and land reform. South Korea's GDP per capita (PPP) rose from approximately $1,200 in 1960 to over $45,000 by 2023. However, the model relied on authoritarian government direction, suppression of labour rights in early decades, and intense work culture with social costs.

5. Inequality

5.1 Income vs Wealth Inequality

Income inequality: Unequal distribution of income (wages, salaries, benefits, investment returns) among a population.

Wealth inequality: Unequal distribution of assets (property, savings, pensions, shares). Wealth inequality is typically far greater than income inequality.

5.2 Measuring Inequality — The Lorenz Curve and Gini Coefficient

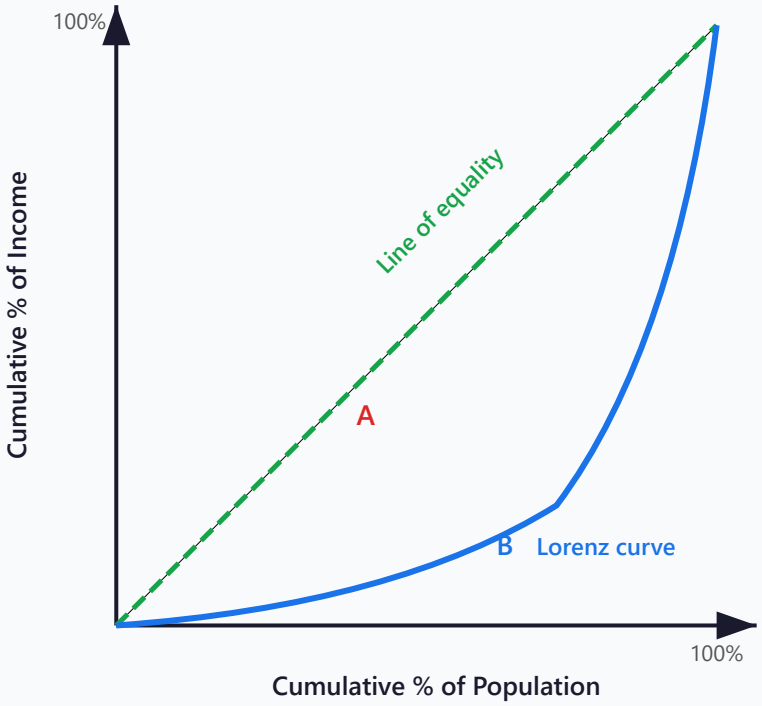

Figure 4: Lorenz Curve — the further the curve bows from the line of equality, the greater the inequality. Gini = A / (A + B)

$$\text{Gini Coefficient} = \text{Area A} / (\text{Area A} + \text{Area B})$$

Gini ranges from **0** (perfect equality — everyone has the same income) to **1** (perfect inequality — one person has all the income). The UK Gini is approximately 0.35; Scandinavian countries are around 0.25-0.28; South Africa is around 0.63.

5.3 Causes of Inequality

- **Wage differentials:** Skills, education, experience, occupation, and bargaining power create large income gaps.

- **Wealth inheritance:** Wealth passed between generations perpetuates inequality.
- **Globalisation:** Benefits skilled workers and capital owners more than unskilled workers.
- **Technological change:** Automation and AI increase demand for skilled workers and reduce demand for routine jobs.
- **Tax and benefit system:** Regressive taxes and inadequate benefits can widen inequality; progressive systems can reduce it.
- **Housing market:** Rising property prices benefit homeowners at the expense of renters, widening the wealth gap.
- **Discrimination:** Gender, racial, and other forms of discrimination create pay gaps.

5.4 Consequences of Inequality

Negative consequences:

- Lower social mobility — entrenched advantage and disadvantage across generations
- Health and social problems (Wilkinson and Pickett's research: more unequal societies have worse health, higher crime, lower trust)
- Political instability and social unrest
- Reduced aggregate demand (low-income groups have higher MPC, so redistributing to them would boost spending)
- Wasted human potential — talented people from disadvantaged backgrounds cannot fulfil their potential

Some inequality may be beneficial:

- Provides incentives to work, innovate, and take risks (entrepreneurship)
- Rewards human capital investment (education, training)
- Extreme equality may reduce incentives and productivity (as seen in some command economies)

5.5 Policies to Reduce Inequality

- **Progressive taxation:** Higher earners pay a larger proportion (UK income tax: 20%, 40%, 45% rates).

- **Transfer payments:** Benefits, pensions, tax credits redistribute income to lower groups.
- **Minimum wage:** National Living Wage (£11.44 from April 2024 for 21+) sets a floor on earnings.
- **Investment in education:** Improving access to quality education increases social mobility.
- **Public services:** Free healthcare (NHS) and education reduce the impact of income inequality on life outcomes.
- **Inheritance tax:** Taxing wealth transfers between generations (currently 40% above £325,000 threshold).

Case Study: Scandinavian Model — Low Inequality and High Development

The Nordic countries (Sweden, Denmark, Norway, Finland) consistently rank among the world's highest for both HDI and equality (Gini coefficients of 0.25-0.28 versus 0.35+ for the UK and 0.39 for the US). Their model combines: highly progressive taxation (top rates of 50-60%), generous universal welfare states (free healthcare, education, parental leave, unemployment insurance), strong trade unions with collective bargaining, heavy investment in education and training, and active labour market policies. The result is high living standards, strong social mobility, low poverty, and high trust — while maintaining competitive, innovative economies. Critics argue the model requires cultural homogeneity and high tax burdens that might not transfer to other contexts, and that it faces challenges from ageing populations and immigration.

Exam tip: Development questions require you to evaluate multiple strategies and consider context. There is no single "correct" development path — what worked for South Korea may not work for Nigeria. Always consider: the country's specific circumstances (resources, institutions, geography), whether the strategy addresses the root barriers to development, short-run vs long-run effects, and potential unintended consequences.

Evaluation — International Economics and Development

- Exchange rate changes have complex, often contradictory effects — appreciation helps consumers but hurts exporters; depreciation is the reverse. Time lags (J-Curve) complicate policy.
- Free trade is theoretically optimal but creates winners and losers — the challenge is compensating losers while capturing efficiency gains.

- Protectionism may be justified in specific cases (infant industry, national security) but is often captured by vested interests and imposes net welfare costs.
- Economic development requires addressing multiple barriers simultaneously — no single policy is sufficient.
- Growth is necessary but not sufficient for development — the distribution of growth matters as much as its quantity.
- The appropriate development strategy depends heavily on a country's specific context — institutions, resources, geography, history, and governance capacity all matter.

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